MAT244 (ODEs) Notes

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1 Introduction to Ordinary Differential Equations

This lecture corresponds to §1.1-1.3 in the textbook.

1.1 What Are Differential Equations?

An **differential equation**, which I will often abreviate as a **DE**, is an equation containing derivatives of an unknown function. The two main types of DEs are **ordinary differential equations**, or **ODEs**, which are equations involving functions depending on a single independent variable, and **partial differential equations**, or **PDEs**, which are equations involving functions which depend on more than one independent variable. For example, the equation:

$$\frac{d^2u}{dt^2} + \frac{du}{dt} + u = f(t)$$

is an ODE that depends on the independent variable t. The equation:

$$-\partial_t^2 u + \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$$

is an example of a PDE that depends on multiple independent variables x, y, z, and t. The focus of this course will be on ODEs.

Example 1.1.1. Consider the following ODE:

$$\frac{dy}{dt} = -y(t) + t$$

How can we determine what this equation represents? From calculus, we know that dy/dt is the derivative of $y(t)$, and that this derivative represents how fast the function y is changing with respect to t. Since we have that $dy/dt = -y(t) + t$, the right hand side shows that the slope will decrease as y increases and will increase as t increases.

Recall that $\left.\frac{dy}{dt}\right|_p$ represents the slope of $y(t)$ passing through the point p. We can thus say that the **direction** of $y(t)$ at point $t = t_0$ is equal to $-y(t_0) + t_0$. With this information, we can plot a **direction field** (or **slope field**) for this differential equation:

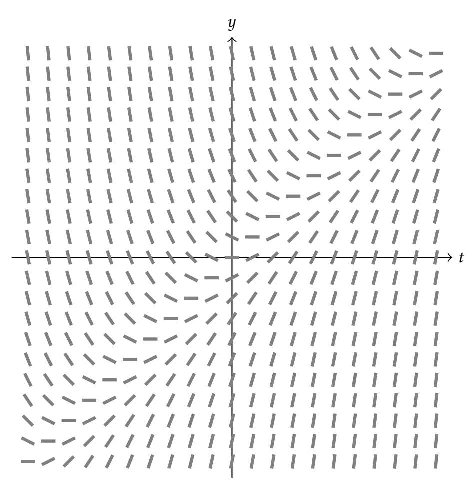

Figure 1: The slope field for the equation $\frac{dy}{dt} = -y(t) + t$

Remark that, in accordance with our intuition, the slope decreases as we move up the y axis and it increases as we move up the t axis. Each line segment in Figure 1 represents the slope of the equation at one of its solutions. Note that if we travel along the line $y(t) = t$, we see that the slope of the solution is always zero. We say that the line $y(t) = t$ is the **equilibrium solution** of the DE and that it does not change with respect to time (i.e. $\frac{dy}{dt} = 0$):

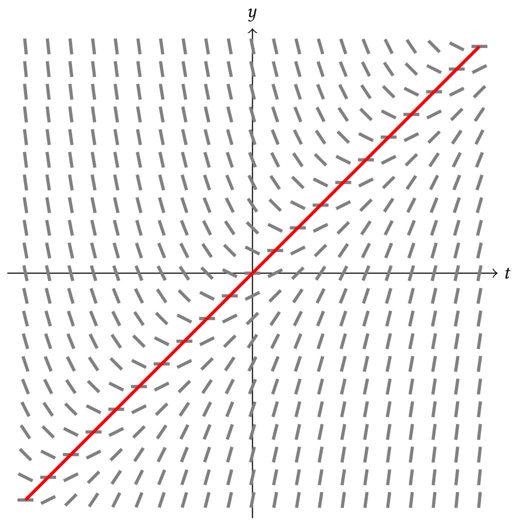

Figure 2: The equilibrium solution (in red) to the equation $\frac{dy}{dt} = -y(t) + t$ is $y(t) = t$.

Example 1.1.2. Consider the differential equation $\frac{dy}{dt} = ay$, where $a \in \mathbb{R}$. Plot direction fields for $a = -1, 0$, and 1. What is/are the equilibrium solution(s)?

Solution. When $a = -1$, we have that $\frac{dy}{dt} = -y$, i.e. the slope of the equation is equal to $-y$. Observe that $\frac{dy}{dt}$ does not change when t changes. The direction field is as follows:

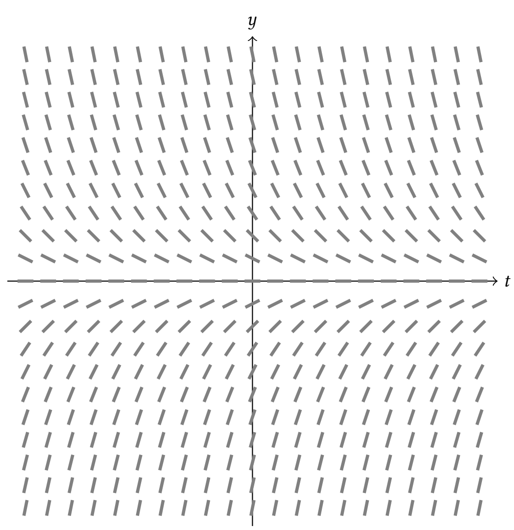

Figure 3: The direction field for $\frac{dy}{dt} = -y$.

Observe that the slope of the equation is equal to the value of $-y$. For example, when $y = 1$, we have that $\frac{dy}{dt} = -1$. Furthermore note that $\frac{dy}{dt}$ is independent of t and hence does not change as t changes. The equilibrium solution for this system is the x-axis $y = 0$.

The direction field for $a = 1$, i.e. $\frac{dy}{dt} = y$, can be plotted in a very similar way.

When $a = 0$, we have that $\frac{dy}{dt} = 0$. Hence the slopes in the direction field will equal zero for all t and y:

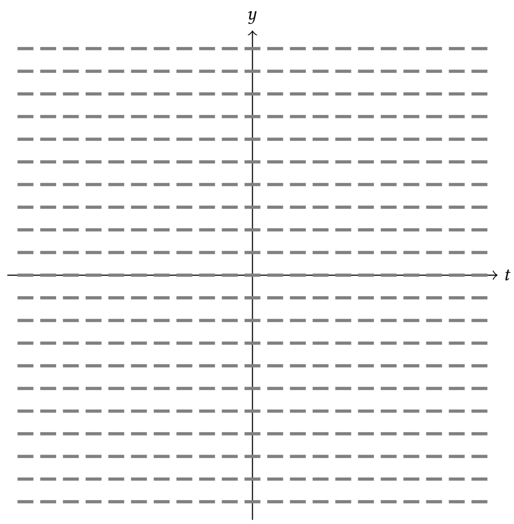

Figure 4: The direction field for $\frac{dy}{dt} = 0$.

Since the slope is always zero, any line passing through the direction field will have slope zero. There are hence infinitely many equilibrium solutions. ✓

1.2 Solutions of Differential Equations

We "solve" differential equations of the form $dy/dt = f(t, y)$ by finding a function $y(t)$ that satisfies it.

Example 1.2.3. Solve the differential equation $\frac{dy}{dx} = 2y(t)$.

Solution. First, rewrite the equation so that all the terms with y appear on one side of the equation and all the terms with t appear on the other:

$$\frac{dy}{dt} = 2y \implies \frac{dy}{y} = 2dt$$

Though it may seem bizarre that we separated dy/dt into two components, we are allowed to do so when solving DEs. We will provide more justified reasoning for doing so in the next lecture.

Now we integrate both sides:

$$\int \frac{dy}{y} = \int 2dt$$
$$\ln|y| = 2t + C$$

Now solve for y:

$$\begin{aligned}\ln|y| &= 2t + C\\ |y| &= e^{2t+C}\\ y(t) &= \pm e^{2t}e^{C}\end{aligned}$$

Note that e^C is a positive constant, so we can replace it with a new constant $A = e^C$ such that:

$$y(t) = Ae^{2t}$$

Lastly let us check our work by integrating $y(t)$ and ensuring it matches the given equation:

$$\begin{aligned}y' &= 2Ae^{2t}\\ y' &= 2y(t)\end{aligned} \qquad \checkmark$$

Suppose that we are also given an initial condition, for instance suppose for the previous example that $y(0) = 1$. This initial condition uniquely determines the value of the constant and hence the function y. With the above example, if we assume $y(0) = 1$, then we find that:

$$\begin{aligned}y(0) = 1 &= Ae^{2(0)}\\ 1 &= A\end{aligned}$$

Example 1.2.4. Solve the ODE $\frac{dy}{dx} = 3y - 2$, given the initial condition $y(0) = 2$.

Solution. First, rearrange the equation so that all the terms with y are on one side and all the terms with x are on the other (remember we are allowed to split $\frac{dy}{dx}$ into components):

$$\frac{1}{3y-2}dy = dx$$

Now integrate both sides:

$$\begin{aligned}\int \frac{1}{3y-2}dy &= \int dx\\ \frac{1}{3}\ln|3y-2| + C_1 &= x + C_2\end{aligned}$$

If we put the constants together we obtain:

$$\begin{aligned}\ln|3y-2| &= 3x + C\\ |3y-2| &= Ae^{3x}\\ 3y - 2 &= \pm Ae^{3x}\\ 3y &= \pm Ae^{3x} + 2\\ y &= \frac{\pm Ae^{3x} + 2}{3}\end{aligned}$$

Now solving for the initial value:

$$
\begin{aligned}
&y(0) = 2 = \frac{\pm Ae^{3(0)} + 2}{3} \\
&2 = \frac{A+2}{3} \\
&6 = A + 2 \implies A = 4
\end{aligned}
$$

So a solution to the initial value problem is $y(t) = \frac{4e^{3t}+2}{3}$. ✓

1.3 Classification of Differential Equations

The **order** of a DE is the highest derivative in the equation. For example, the order of $y' = y(t)$ is 1, and the order of $y'' + y + t = 0$ is 2. The general form of an order n DE is:

$$F(x, u(x), u'(x), u''(x), \ldots, u^{(n)}(x)) = 0$$

A solution of the ODE $y^{(n)} = f(x, y, y', \ldots, y^{(n-1)})$ on the interval $a < x < b$ is a function ϕ such that $\phi', \phi'', \ldots, \phi^{(n)}$ exists and satisfies $\phi^{(n)} = f(x, \phi, \phi', \ldots, \phi^{(n-1)})$ for all $x \in (a, b)$. Note that ϕ and all necessary derivatives must exist on $a < x < b$ and ϕ must satisfy the differential equation.

For example, consider $y'' + y = 0$, or $y'' = -y$. Solutions to this DE will be of the form e^{ix} or make use of sines and cosines (for instance, it is easy to check that $y = \sin(x) + \cos(x)$ is a solution). However, naturally the question arises: does a solution always exist, if so it is unique? And how do we find it? We will learn how to answer those questions in the next lecture.

2 Linear & Nonlinear ODEs, Integrating Factors

This lecture corresponds to §1.3 and §2.1 in the textbook.

2.1 Linear & Nonlinear ODEs

Definition 2.1.1

We say a differential equation $F(x, y, y', y'', \ldots, y^{(n)}) = 0$ is **linear** if F is a linear function of $y, y', y'', \ldots, y^{(n)}$. Thus the general form of a linear differential equation is:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

We say an equation is **nonlinear** if it is not linear.

The equation $y + y'' = 0$ is an example of a linear DE, as it can be written in the form $a_0(x)y + a_2(x)y'' = 0$ where $a_0(x) = a_2(x) = 1$. Another example of a linear DE is $y' \sin(x) = x$, where $a_1(x) = \sin(x)$ and $g(x) = x$. Conversely, the equation $y' + y^2 = 0$ is a nonlinear DE as it cannot be written in a linear form.

In mathematics, linear theory has been very well developed, but nonlinear theory has not been developed to the same degree. However, we have been able to approximate nonlinear equations by linear ones using **linear approximations**.

For example, consider the following differential equation representing an oscillating pendulum:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0$$

This equation is obviously nonlinear due to the $\sin\theta$. However, when the angle θ is small, we can approximate $\sin\theta \approx \theta$, hence we can linearize the equation as:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0$$

which is an equation we can solve (see exercises).

2.2 Integrating Factors with Linear First Order ODEs

In this section, we will be working with equations of the form $\frac{dy}{dx} + p(x)y = g(x)$, where p and g are functions of the independent variable t. We ,introduce a new method of solutions called integrating factors, with $\mu(t)$ being the integrating factor.
Example 2.2.2. Solve the differential equation $y' + 2y = 1$ using integrating factors.

Solution. The first step in solving an ODE with integrating factors is to multiply the entire equation by a function $\mu(t)$ which we will choose so the entire function is readily integrable. We have:

$$\mu(t)\frac{dy}{dt} + \mu(t)2y = \mu(t) \tag{1}$$

The second step is to try and find the value of $\mu(t)$ such that this equation becomes readily integrable. We do this by considering the left side of Equation (1) as the derivative of some recognizable expression. Observe that the term $\mu(t)\frac{dy}{dx}$ is part of the differentiating product $\mu(t)y$. Indeed if we recall the product rule:

$$\frac{d}{dt}\left[\mu(t)y\right] = \mu(t)\frac{dy}{dt} + \frac{d\mu}{dt}y$$

we can find the integrating factor $\mu(t)$ by setting $\mu(t)2y = \frac{d\mu}{dt}y$. Isolating $\mu(t)$ we obtain:

$$\begin{aligned}
&\frac{1}{\mu(t)}d\mu = 2dt \\
\implies &\int \frac{1}{\mu(t)}d\mu = \int 2dt \\
\implies &\ln|\mu(t)| = 2t + C \\
\implies &\mu(t) = e^{2t+C} \\
\implies &\mu(t) = Ae^{2t}
\end{aligned}$$

Hence Equation (1) becomes (we can cancel out the constant A):

$$e^{2t}\frac{dy}{dt} + e^{2t}2y = e^{2t} \tag{2}$$

We remark that now the left side of Equation (2) is obviously the derivative of the product $e^{2t}y$, so we can simplify the leftside furthermore to obtain:

$$\frac{d}{dt}\left[e^{2t}y\right] = e^{2}t$$

The third step is to integrate both sides of the Equation (3), which we can now do with the chosen integrating factors, and isolate y. Remark that on the left side of the equation the derivative and integral cancel each other out:

$$\begin{aligned}
&\cancel{\int}\frac{\cancel{d}}{dt}\left[e^{2t}y\right]\cancel{dt} = \int e^{2t}dt \\
\implies &e^{2t}y = \frac{e^{2}t}{2} + C \\
\implies &y = \frac{1}{2} + Ce^{-2t}
\end{aligned}$$

✓

Remark 2.2.3. For ODEs of the form $\frac{dy}{dt} + p(t)y = g(t)$, the integrating factor is always of the form $\mu(t) = \exp\int p(t)dt$.

Example 2.2.4. Solve the differential equation $y' + 2ty = e^{-t^2}$ using integrating factors.

Solution. We begin by multiplying the equation by the integrating factor $\mu(t)$:

$$\mu(t)y' + \mu(t)2ty = \mu(t)e^{-t^2} \tag{3}$$

We first note that $p(t) = 2t$ and $g(t) = e^{-t^2}$. Using the remark above, we find that:

$$\begin{aligned} &\mu(t) = \exp \int 2t dt \\ \iff\ &\mu(t) = e^{t^2+C} \\ \iff\ &\mu(t) = Ae^{t^2} \end{aligned}$$

So Equation (3) becomes:

$$e^{t^2}\frac{dy}{dt} + 2te^{t^2}y = 1$$

We recognize the left side of the equation as the derivative of the product $\mu(t)y$. Integrating both sides of the equation yields:

$$\begin{aligned} &\cancel{\int} \frac{\cancel{d}}{dt}\left[e^{t^2}y\right]\cancel{dt} = \int dt \\ \implies\ &e^{t^2}y = t + C \\ \implies\ &y = e^{-t^2}(t + C) \qquad \checkmark \end{aligned}$$

3 Integrating Factors & Separable Equations

3.1 More Examples of Integrating Factors

Example 3.1.1. Solve the differential equation $\frac{dy}{dt} - y = e^{-t}$.

Solution. Start by multiplying the equation by the integrating factor $\mu(t)$:

$$\mu(t)\frac{dy}{dt} - \mu(t)y = \mu(t)e^{-t} \tag{4}$$

Using Remark 2.2.3. (remember this), we find the integrating factor. Let $p(t) = -1$:

$$\begin{aligned} \mu(t) &= \exp \int -dt \\ \iff \mu(t) &= e^{-t+C} \\ \iff \mu(t) &= Ae^{-t} \end{aligned}$$

Equation (4) becomes:

$$e^{-t}\frac{dy}{dt} - e^{-t}y = e^{-2t} \tag{5}$$

We recognize the left side of equation (5) as the derivative of the product $e^{-t}y$. Integrating both sides, we obtain:

$$\begin{aligned} \cancel{\int} \frac{\cancel{d}}{dt}\left[e^{-t}y\right] \cancel{dt} &= \int e^{-2t}dt \\ \implies e^{-t}y &= -\frac{e^{-}2t}{2} + C \\ \implies y &= -\frac{e^{-t}}{2} + Ce^{t} \end{aligned}$$

✓

Example 3.1.2. Solve the differential equation $\frac{dy}{dt} = y\sin(t) + 2te^{-\cos(t)}$.

Solution. Let us first rearrange the equation to put all terms with y on the left side, then multiply by the integrating factor $\mu(t)$:

$$\mu(t)\frac{dy}{dt} - \mu(t)y\sin(t) = \mu(t)2te^{-\cos(t)} \tag{6}$$

We recall Remark 2.2.3 (again, remember this) to find $\mu(t)$. Let $p(t) = -\sin(t)$:

$$\begin{aligned} \mu(t) &= \exp \int -\sin(t)dt \\ &= e^{\cos(t)+C} \\ &= Ae^{\cos(t)} \end{aligned}$$

Plugging our newfound value of $\mu(t)$ into Equation (6):

$$e^{\cos(t)}\frac{dy}{dt} - e^{\cos(t)}y\sin(t) = 2t$$

We recognize the left side of the equation as the derivative of the product $e^{\cos(t)}y$. Integrating and isolating for y yields:

$$\begin{aligned} & \cancel{\int}\frac{\cancel{d}}{dt}\left[e^{\cos(t)}y\right]\cancel{dt} = \int 2t\,dt \\ \iff & e^{\cos(t)}y = t^2 + C \\ \iff & y = e^{-\cos(t)}(t^2 + C) \end{aligned} \qquad \checkmark$$

3.2 *Separable Equations*

We learned how to use integrating factors to solve differential equations of the form $\frac{dy}{dt} + p(x)y = g(x)$. We will now shift our focus to differential equations of the form $\frac{dy}{dt} = f(y)g(t)$. We say this class of differential equations are **separable**, and we can solve them by putting like variables on the same sides of the equality, and then integrating both sides.

Example 3.2.3 (Newton's Law of Cooling)**.** Consider the following ODE:

$$\frac{dB}{dt} = K(E - B)$$

where $E, K \in \mathbb{R}$, with the initial condition $B(0) = B_0$. To solve this equation, we begin by putting variables with B on the left side, and variables with t (there are none) on the right side:

$$\frac{dB}{E - B} = K\,dt$$

Integrating both sides yields:

$$\begin{aligned} & \int \frac{dB}{E - B} = \int K\,dt \\ \implies & -\ln|E - B| = Kt + C \\ \implies & \ln|E - B| = -Kt - C \\ \implies & E - B = \pm e^{-Kt-C} \\ \implies & -B = Ae^{-Kt} - E \\ \implies & B = E - Ae^{-Kt} \end{aligned}$$

where $A = \pm e^{-C}$. We have that $B(0) = B_0$, so we can find the value of A given the initial value:

$$\begin{aligned} & B_0 = E - A \\ \implies & A = E - B_0 \end{aligned}$$

Hence the solution to the ODE is:

$$B(t) = E - (E - B_0)e^{-Kt}$$

Example 3.2.4. Solve the differential equation $\frac{dy}{dt} = 6y^2t$, given the initial condition $y(1) = \frac{1}{3}$. Additionally, find the interval of valid solutions for t.

Solution. We will solve this equation by separation. First, bring all terms with y on the left side and all with t on the right side:

$$\frac{dy}{y^2} = 6tdt$$

Integrating both sides of the equation yields:

$$\begin{aligned} \int \frac{dy}{y^2} &= \int 6tdt \\ \implies \frac{-1}{y} &= 3t^2 + C \\ \implies y &= \frac{-1}{3t^2 + C} \end{aligned}$$

Now, find C using the initial value $y(1) = \frac{1}{3}$:

$$\begin{aligned} \frac{1}{3} &= \frac{-1}{3+C} \\ \implies 3 &= -3 - C \\ \implies C &= -6 \end{aligned}$$

So the solution to the equation is:

$$y = \frac{-1}{3t^2 - 6}$$

Now to find valid solutions for t, we need to find the restrictions on the solution we found above. We know that $3t^2 - 6 \neq 0 \implies 3t^2 \neq 6 \implies t \neq \pm\sqrt{2}$. So the interval of valid solutions for t is:

$$t = \{t \in \mathbb{R} : t \neq \pm\sqrt{2}\}$$

This implies that the solutions are discontinuous at points $t = -\sqrt{2}$ and $t = \sqrt{2}$. ✓

Example 3.2.5. Solve the differential equation $\frac{dy}{dt} = \frac{ty^3}{1+t^2}$ with the initial condition $y(0) = 1$.

Solution. We will solve by separation. We first bring all terms with y on the left side and all with t on the right:

$$\frac{dy}{y^3} = \frac{t}{1+t^2}dt$$

Integrating both sides yields:

$$\begin{aligned}
&\int \frac{dy}{y^3} = \int \frac{t}{1+t^2}dt \\
\implies &\frac{-1}{2y^2} = \frac{1}{2}\ln|1+t^2| + C \\
\implies &y^{-2} = -\ln|1+t^2| - 2C \\
\implies &y^2 = \frac{1}{-\ln|1+t^2| - 2C} \\
\implies &y = \pm\frac{1}{\sqrt{-\ln|1+t^2| - 2C}}
\end{aligned}$$

Using the intial condition $y(0) = 1$ we solve for C:

$$\begin{aligned}
&1 = \pm\frac{1}{\sqrt{-2C}} \\
\implies &\pm\sqrt{-2C} = 1 \\
\implies &-2C = 1 \\
\implies &C = \frac{-1}{2}
\end{aligned}$$

So the solution to the equation is:

$$y = \frac{1}{\sqrt{1 - \ln|1+t^2|}} \qquad \checkmark$$

Example 3.2.6. Solve the differential equation $\frac{dy}{dt} = e^{y-t}\sec(y)(1+t^2)$ with the intiial condition $y(0) = 0$.

Solution. Again, we proceed by separating y and t:

$$\begin{aligned}
&\frac{dy}{e^y \sec(y)} = e^{-t}(1+t^2)dt \\
\iff &e^{-y}\cos(y)dy = e^{-t}(1+t^2)dt
\end{aligned}$$

Integrating both sides can be done by using integration by parts on both integrals, though for brevity I will omit those calculations:

$$\begin{aligned}
&\int e^{-y}\cos(y)dy = \int e^{-t}(1+t^2)dt \\
\implies &\frac{e^{-y}(\sin(y) - \cos(y))}{2} = e^{-t}(t^2+2t+3) + C \\
\implies &e^{-y}(\sin(y) - \cos(y)) = 2e^{-t}(t^2+2t+3) + 2C
\end{aligned}$$

This is unfortunately the best we can simplify the equation, but we can still solve for C using the condition $y(0) = 0$:

$$\begin{aligned}
&1 = 6 + 2C \\
\implies &C = \frac{-5}{2}
\end{aligned}$$

So an implicit solution to the differential equation is:

$$e^{-y}(\sin(y) - \cos(y)) = 2e^{-t}(t^2 + 2t + 3) - 5$$

✓

4 Modelling First Order Differential Equations & Autonomous Differential Equations

4.1 Modelling

When modelling differential equations there are three important steps to follow:

1. **Translation**: this step involves understanding the problem, assigning variables to unknown quantities and developing equation(s) that describe the physical phenomena.
2. **Solving**: here, we decide whether to solve the system numerically or analytically, approximately or exactly, etc.. Our goal here is to bring all information about the problem to bear.
3. **Interpret**: ensure that your answer you found in the previous step makes sense. Is your solution unique? Is it stable?

Example 4.1.1 (Radioactive Decay)**.** Consider the following differential equation, which represents decay of a radioactive isotope:

$$\frac{dN}{dt} = -\lambda N(t)$$

In this example, $N(t)$ is the number of the radioactive isotope and $\lambda > 0$ is the decay constant. This equation can easily be solved using separation:

$$\begin{aligned}
&\frac{dN}{N(t)} = -\lambda dt \\
\implies &\int \frac{dN}{N(t)} = \int -\lambda dt \\
\implies &\ln |N(t)| = -\lambda t + C \\
\implies &N(t) = Ae^{\lambda t}
\end{aligned}$$

We remark that as $t \to \infty$, $N(t) \to 0$, which makes sense as the isotope is decaying.

Example 4.1.2 (Newton's Law of Cooling)**.** Suppose that we immerse a body in an environment with constant temperature E. Let $B(t)$ be the function representing the temperature of the body. We can represent the heat conductivity of the object with the equation:

$$\frac{dB}{dt} = K(E - B(t))$$

where $K > 0$ represents how well the body's material conducts heat. We can solve this equation by separation, which gives:

$$B(t) = E - \frac{E - B_0}{e^{Kt}}$$

Example 4.1.3 (Falling Bodies)**.** We represent the fall of a body with mass m with net forces $F(t, v)$ and acceleration $\frac{dv}{dt}$ using the differential equation:

$$m\frac{dv}{dt} = F(t, v)$$

For this class assume that downward displacements and forces are negative. Assume also that the force of gravity, $F_g = -mg$ where $g = 10m/s^2$.

Additionally, we have kinematic equations for acceleratio $\mathbf{a}$, velocity $\mathbf{v}$, and displacement $\mathbf{r}$:

$$\begin{aligned}
\mathbf{a} &= \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2} \\
\mathbf{v} &= \int \mathbf{a}dt = \mathbf{a}t + v_0 \\
\mathbf{r} &= \int \mathbf{a}t + v_0 dt = \frac{1}{2}\mathbf{a}t^2 + v_0 t + r_0
\end{aligned}$$

We will use these equations to solve the following exercise:

Example 4.1.4. A 50kg object is shot from a cannon straight up with an initial velocity of $10m/s$ off the very tip of a bridge. If the air resistance is given by $5\mathbf{v}$ determine the velocity of the object at any time t.

Solution. We have that:

$$m\frac{dv}{dt} = -10m - 5v$$

Which can be written as:

$$\frac{dv}{dt} + \frac{5}{m}v = -10$$

Let us solve this system using integrating factors. Let $\mu(t) = \exp \int p(t) \implies \mu(t) = \exp\left(\frac{5}{m}t\right)$. Multiplying the equation by the integrating factor gives: ✓

$$\begin{aligned}
& e^{5t/m}\frac{dv}{dt} + \frac{5}{m}e^{5t/m}v = -10e^{5t/m} \\
\iff & \frac{d}{dt}\left[e^{5t/m}v\right] = -10e^{5t/m} \\
\iff & e^{5t/m}v = \int -10e^{5t/m}dt \\
\iff & v = -2m + Ce^{-5t/m}
\end{aligned}$$

Since we know $v(0) = 0$ and $m = 50$, we have:

$$\begin{aligned}
& v(0) = 0 = -100 + C \\
\implies & C = 100
\end{aligned}$$

So the velocity of the object at time t is:

$$v(t) = -100 + 100e^{-t/10}$$

4.2 Autonomous Equations

Definition 4.2.5

We say a first order DE is **autonomous** if the independent variable does not appear explicitly in the equation. For example, the ODE $\frac{dy}{dt} = f(y)$ is autonomous as it depends on y alone.

We have already seen some examples of autonomous ODEs in the first lecture, we will now classify certain two main types of autonomous ODEs:

Definition 4.2.6

We say an autonomous ODE is **exponential** if it can be written in the form $\frac{dy}{dt} = ry$, where $r \neq 0$. If r is positive, we say the equation represents **exponential growth**, whereas if it is negative, the equation represents **exponential decay**.

Now let us modify the exponential growth equation $\frac{dy}{dt} = ry$ such that $\frac{dy}{dt} \approx ry$ when y is small, and $\frac{dy}{dt} \longrightarrow 0$ as $y \longrightarrow K$, where K is the population that can be sustained. We write the equation as:

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right)$$

where the term $\left(1 - \frac{y}{K}\right)$ is close to 1 when y is small, and is close to 0 when $y \longrightarrow K$.

Definition 4.2.7

We say an autonomous ODE is **logistic** if it can be written in the form $\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right)y$, where $r \neq 0$, $K = r/a$. If r is positive, we say the equation represents **logistic growth**, whereas if it is negative, the equation represents **logistic decay**.

We say r is the **intrinsic growth rate** of the system and K is the system's **saturation level** or **carrying capacity**.

Observe that for small y, the rate of change is close to 1, but as y tends to k, the rate of change approaches 0.

Remark 4.2.8. Solving the logistic equation (by using separation of variables) yields the following general solution:

$$y(t) = \frac{Ky_0}{y_0 + (K - y_0)e^{-rt}}$$

Where y_0 is the value of $y(0)$. Observe that as $t \longrightarrow \infty$, $y \longrightarrow \frac{Ky_)}{y_0} = K$.

5 Linear and Non-Linear Equations and Exact Differential Equations

5.1 The Existence and Uniqueness Theorem

Consider a lineaer differential equation. We have the following theorem:

Theorem 5.1.1

Let $y' + p(x)y = g(x)$ be a linear differential equation with $p(x)$ and $g(x)$ being C^0 functions on an open interval $I\colon a < x < b$ containing the point $x = x_0$. Then, there exists a unique function $\phi(x)$ that satisfies the system for each $x \in I$, and that also satisfies the initial condition $y(x_0) = y_0$ where y_0 is an arbitrary initial value.

Proof. Coming soon. ■

If the differential equation is non linear, of the form $y' = f(x, y)$ with initial value $y(x_0) = y_0$, we have an analogous theorem:

Theorem 5.1.2

Let f and $\frac{\partial f}{\partial y} = \partial_y f = f_y$ be continuous on some open rectangle $a < x < b$ and $c < y < d$ containing the point $x_0, y_0)$. Then, in some interval $x_0 - h < x < x_0 + h$ contained in $a < x < b$, there exists a unique solution $y = \phi(x)$ to the initial value problem.

Proof. Coming soon. ■

Example 5.1.3. Consider the following first order differential equation:

$$\frac{dy}{dx} = \frac{1 + x^2}{3y - y^2}$$

Does this equation satisfy the conditions of Theorem 5.1.2?

Solution. First let us consider the function $f(x, y) = (1 + x^2)/(3y - y^2)$. This is continuous if and only if $3y - y^2 \neq 0 \iff y \neq 0$ and $y \neq 3$. The partial derivative with respect to y of f is $f_y = -(1 + x^2)(3 - 2y)/(3y - y^2)$ which is again continuous if and only if $y \neq 0$ and $y \neq 3$. Hence, the existence and uniqueness theorem holds on any rectangle that does not meet the lines $y = 0$ or $y = 3$. In particular, if we consider an arbitrary initial point (x_0, y_0) such that $y_0 \neq 0, 3$, we can find an $h > 0$ such that the IVP has a unique solution on the interval $x_0 - h < x < x_0 + h$. ✓

5.2 Exact Differential Equations, Part I

Consider the following first order ODE of the form:

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

If this equation were linear, then we can rewrite it in the form $M(x, y) = M_1(x)y - M_2(x)$ and then solve using integrating factors. However, what do we do when the equation is nonlinear? We can try a different approach if the function is exact.

Definition 5.2.4

We say the differential equation $M(x, y) + N(x, y)\frac{dy}{dx} = 0$ is **exact** if $M_y(x, y) = N_(x, y)$, that is the partial derivative of M with respect to y equals the partial derivative of N with repsect to x.

For example, the differential equation $(2x + 3) + (2y - 2)y' = 0$ is exact, as if we let $M = 2x + 3$, $N = 2y - 2$, we see that $M_y = N_x = 0$. However, the equation $(2x + 4y) + (2x - 2y)y' = 0$ is not exact, as if we let $M = 2x + 4y$ and $N = 2x - 2y$, we see that $M_y = 4$ but $N = 2$.

Naturally, the question arises, how do we solve an exact equation? Let us first define the following derivative:

Definition 5.2.5

Let $\psi : \mathbb{R}^n \to \mathbb{R}$ be differentiable with variables $x_1, x_2, \ldots, x_n$. We say that the **total derivative** of ψ with respect to some parameter t is given by:

$$\frac{d\psi}{dt} = \sum_{i=1}^{n} \frac{\partial\psi}{\partial x_i}\frac{dx_i}{dt}$$

In this class, we will only consider $\psi(x, y)$ and its total derivative with respect to x, which is equal to:

$$\frac{d\psi}{dt} = \frac{\partial\psi}{\partial x}\frac{dx}{dx} + \frac{\partial\psi}{\partial x}\frac{dy}{dx} = \frac{\partial\psi}{\partial x} + \frac{\partial\psi}{\partial x}\frac{dy}{dx}$$

If we can find a function $\psi(x, y)$ such that $\frac{\partial\psi}{\partial x} + \frac{\partial\psi}{\partial x}\frac{dy}{dx} = M(x, y) + N(x, y)\frac{dy}{dx} = 0$, then we can write $\frac{\partial\psi}{\partial x} = 0 \implies \psi(x, y) = c$ and we will have solved the system implicitly for y.

Our task is hence to find the equation $\psi(x, y)$ such that $\frac{\partial\psi}{\partial x} = M(x, y)$ and $\frac{\partial\psi}{\partial y} = N(x, y)$.

Observe that:

$$\frac{\partial\psi}{\partial x} = M(x, y) \implies \psi(x, y) = \int M(x, y)dx + h(y)$$

If we now differentiate $\psi(x, y)$ with respect to y, we obtain:

$$\frac{\partial\psi}{\partial y} = N(x, y) = \int M_y dx + h'(y) \implies h'(y) = N(x, y) - \int M_y dx$$

This says that the right hand side of the equation must only be a function of y, so the partial derivative of the right hand side with respect to x must be zero. If this is true, then we only have to find $h(y)$ (by simple integration) and we obtain the implicit solution:

$$\psi(x, y) = \int M(x, y)dx + h(y) = c$$

This method of solving ODEs may seem complicated. We will do more examples next lecture.